ASSIGNMENT-7: DICTIONARIES

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GROUP-E1, ROLL NO:7

1. Create a dictionary containing the names of the winners of a competition as keys and number of their wins as values.

In [1]:
```
winners={'Jack':5, 'Michael':8, 'Fiona':5}
print(winners)
```

```
{'Jack': 5, 'Michael': 8, 'Fiona': 5}
```

10. Create a nested dictionary containing the employee ID, name and designation of three employees and the then add a fourth employee as a new dictionary with his 1name, designation and salary. Display the old and new dictionary and finally remove the third employee and display the collection.

In [5]:
```
emp={1:{'Empid': 52652, 'Name': 'Srabani', 'Desgn': 'Manager'},
     2:{'Empid': 52656, 'Name': 'Sonia', 'Desgn': 'Accountant'},
     3:{'Empid': 52651, 'Name': 'Sam', 'Desgn': 'Deputy Manager'}
    }
d={4:{'Empid': 52653, 'Name': 'Sejal', 'Desgn': 'Clerk'}}
emp.update(d)
emp.pop(3)
print(emp)
```

```
{1: {'Empid': 52652, 'Name': 'Srabani', 'Desgn': 'Manager'}, 2: {'Empid': 52656, 'Name': 'S
onia', 'Desgn': 'Accountant'}, 4: {'Empid': 52653, 'Name': 'Sejal', 'Desgn': 'Clerk'}}
```

11. Count and display frequency of a word that appears in a given string. [Hint: take an empty dictionary, use split() to identify each words of string.]

In [7]:
```
d={}
s=input('Enter String: ')
l=s.split()
st=set(l)
for i in st:
    d[i]=l.count(i)
print(d)
```

```
Enter String: hello hello hello i am am srabani
```

Out[7]:
```
{'srabani': 1, 'am': 2, 'i': 1, 'hello': 3}
```

12. Create a dictionary whose keys are month names and whose values are the number of days in the corresponding months.

a) Ask the user to enter a month name and use the dictionary to tell how many days are there in that month.

b) Print all of the keys in alphabetical order.

c) Print all of the month with 31 days.

In [16]:
```
months={'January':31, 'February':28, 'March':31, 'April':30, 'May':31, 'June':30, 'July':31
```

```
#(a)
m=input('Enter month: ')
print(f'{m.title()} has {months.get(m.title())} days.')

#(b)
print('Keys in alphabetical Order: ', sorted(months.keys()))

#(c)
m31=[]
for i in months:
    if months[i]==31:
        m31.append(i)
print('Months with 31 days are: ', m31 )
```

```
Enter month: july
July has 31 days.
Keys in alphabetical Order:  ['April', 'August', 'December', 'February', 'January', 'July',
'June', 'March', 'May', 'November', 'October', 'September']
Months with 31 days are:  ['January', 'March', 'May', 'July', 'August', 'October', 'Decembe
r']
```

13. Repeatedly ask the user to enter a team name and how many games the team has won and how many they lost. Store this information in a dictionary whose keys are the team names and the values are list of the form [wins, losses]. Perform the following using this dictionary:

a) Enter any team name and print the team's winning percentage.

b) Create a list whose entries are the number of wins of each team.

In [22]:
```
d={}
ans='y'
while(ans=='y'):
    name=input('Enter team name:')
    wins=int(input('Enter number of wins:'))
    losses=int(input('Enter number of losses:'))
    l=[wins, losses]
    d[name]=l
    ans=input('Anymore?(y/n):')

#(a)
t=input('Enter team name: ')
l=d.get(t)
win_pt=l[0]/sum(l) * 100
print(f'Wins % of team {t} is {win_pt}%')

#(b)
wins_list=[i[0] for i in d.values()] #(id.values())[0]
print('List of wins: ', wins_list)
```

```
Enter team name:titanic
Enter number of wins:7
Enter number of losses:6
Anymore?(y/n):y
Enter team name:losers
Enter number of wins:7
Enter number of losses:1
Anymore?(y/n):y
Enter team name:players
Enter number of wins:7
Enter number of losses:5
Anymore?(y/n):n
Enter team name: losers
Wins % of team losers is 87.5%
List of wins:  [7, 7, 7]
```